PCI PCL-AI · COURSE OUTLINE

Finance and delivery. One profession.

The syllabus for the AI Project Controls Leader — the credential that examines the accounting **and** the arithmetic of delivery, because a project needs both and almost nobody is trained in both.

13 domains · 61 knowledge areas →

WHY THIS CREDENTIAL EXISTS

Two professions.
One project.

THE ACCOUNTANT

Knows the money

- When revenue may be recognised
- What a provision must satisfy
- Which costs may be capitalised

Rarely examined on a critical path, an earning rule, or why a schedule slip becomes a cost.

THE ENGINEER

Knows the work

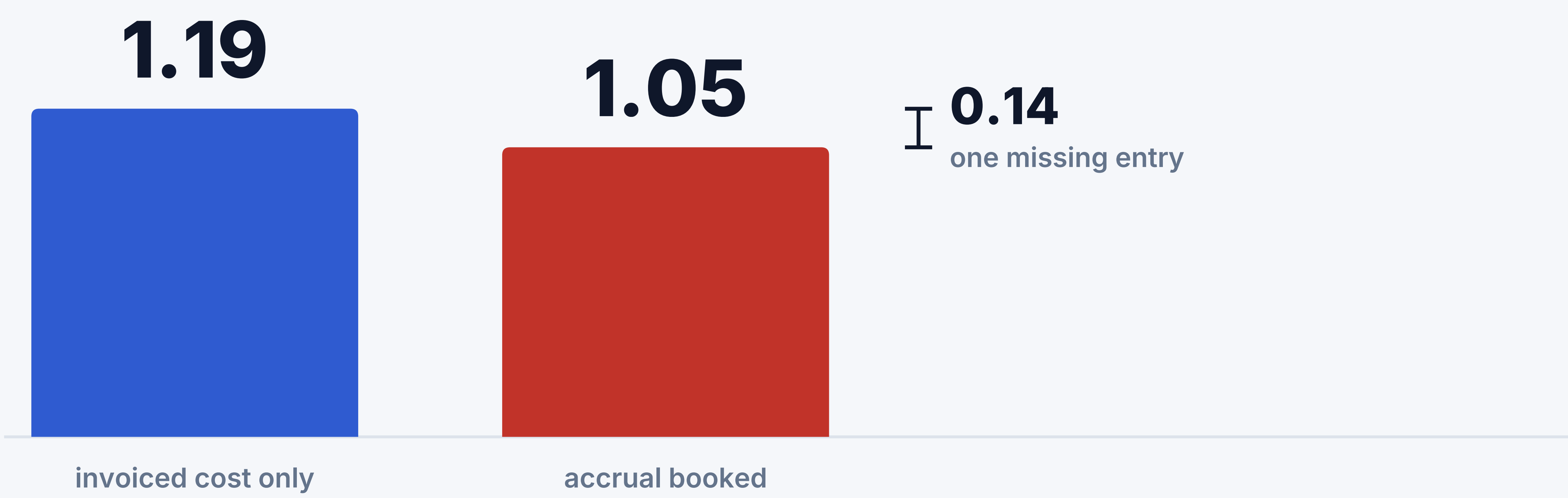
- Where the critical path runs
- What the float really is
- How progress is measured

Rarely examined on cut-off, a contract asset, or the difference between billed and earned.

A project lives in the overlap. **The overlap is where the money is lost**, and it is the one place neither training looks.

WHAT THE OVERLAP COSTS

A CPI of 1.19 and 1.05 from the same month



Earned value 2,200,000. Invoiced cost 1,850,000. The error is **accounting** — a cut-off failure. The damage is **delivery** — a performance index a board will act on.

> An accountant who never reads a CPI will not see it. A planner who never raises an accrual will not see it either. **That is the case for this credential in one month-end.**

THE SYLLABUS

Forty, forty, twenty

40%
Finance, accounting and reporting

40%
Project management

20%
Governed AI

13
domains

61
knowledge areas

26
sector case studies

THE FINANCE HALF

Standards a controls lead is examined on

IFRS 15	revenue from contracts with customers — the five-step model, over-time recognition, variable consideration and the constraint
IAS 37	provisions and contingent liabilities — behind every warranty reserve, onerous-contract charge and dispute disclosure
IAS 2	the materials and work-in-progress a project holds in store
IAS 16 / IFRS 16	the plant a project owns, and the plant and premises it leases
IAS 23	the financing cost of building a major asset
IAS 1	presentation of financial statements
IAS 11	the construction-contracts standard IFRS 15 replaced — covered so legacy terminology in an older file still reads

DOMAIN 2 · REVENUE

The recognition chain, in full

Percentage of completion (PoC) = Costs incurred to date / Total estimated costs

over-time recognition by the input method

Cumulative revenue = PoC × Transaction price

revenue earned to date

Period revenue = Cumulative revenue − revenue recognised in prior periods

what lands in this month

Contract asset (liability) = cumulative revenue − cumulative billed

over- or under-billing, on the balance sheet

> Recognition is not billing. A project can be profitable, fully billed and **still carry a contract liability**.

DOMAINS 1–2 · MEASUREMENT

Before the number exists

Assets = Liabilities + Equity

the identity every ledger satisfies

catch-up = PoC × revised price − revenue recognised to date

the cumulative catch-up when the price is revised

Revised transaction price = fixed price + constrained variable consideration

variable consideration, constrained

Capitalised borrowing cost = weighted-average qualifying expenditure × capitalisation rate

IAS 23 — financing cost inside the asset

NRV = estimated selling price − costs to complete and sell

IAS 2 — the write-down test on stored materials

DOMAIN 7 · COMMERCIAL

Contract to cash

Amount = quantity × rate

the bill-of-quantities line, before anything else happens

amount due = gross value − retention − previous payments

the payment application

Fee = target fee + contractor's share × (target cost − actual cost)

target-cost pain/gain share

LD exposure = LD rate × forecast days late

liquidated damages, forecast not incurred

CCC = DSO + DIO − DPO

cash conversion cycle, in days

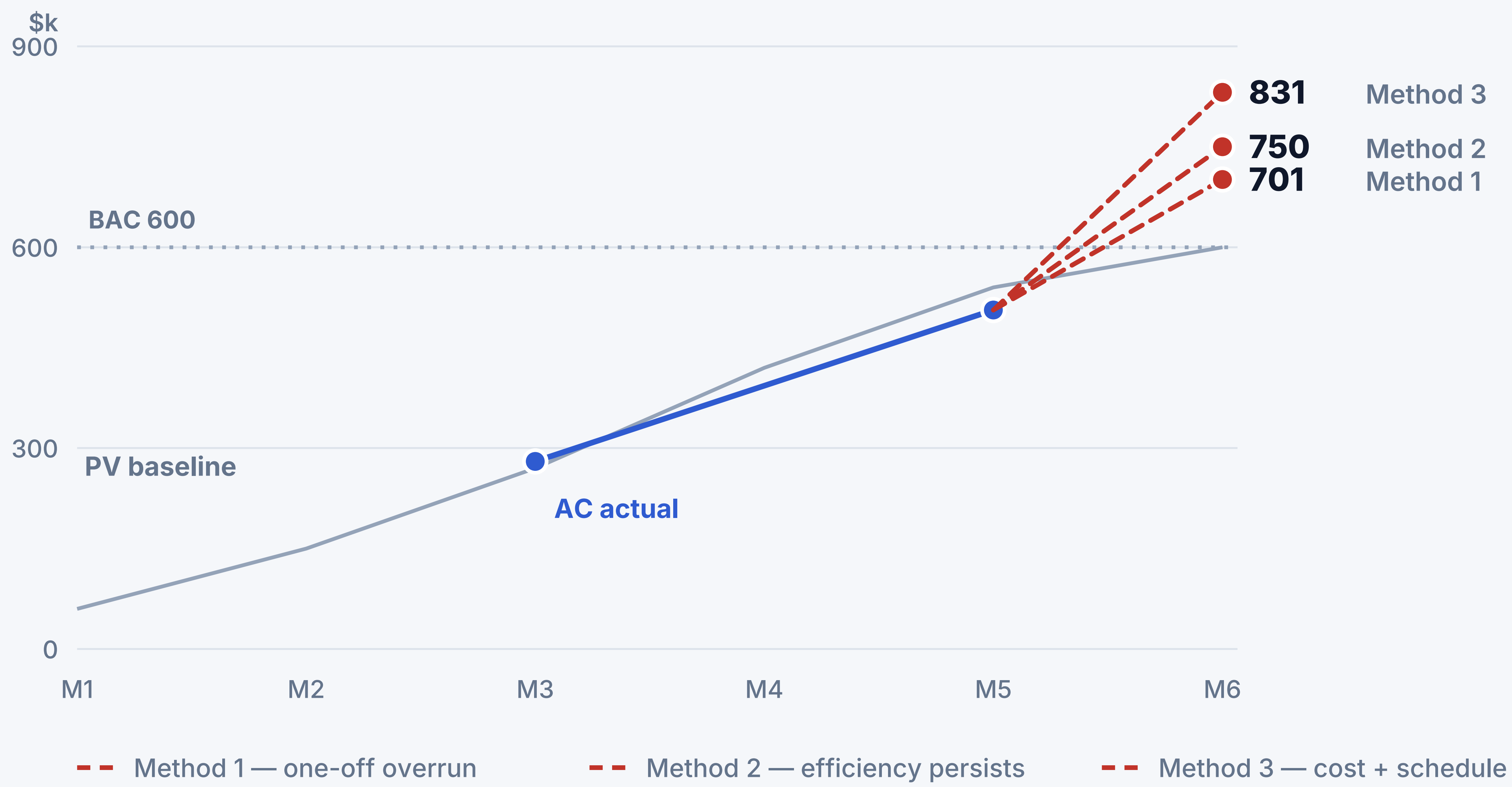
DOMAIN 6 · EARNED VALUE

The delivery half

CV = EV − AC	cost variance
SV = EV − PV	schedule variance, in money
CPI = EV / AC	cost performance index
SPI = EV / PV	schedule performance index
SV(t) = ES − AT	earned schedule variance, in time
SPI(t) = ES / AT	the index that still works near completion

DOMAIN 6 · THE HEART OF IT

There is no 'the' EAC



Three formula-valid forecasts, **130,000 apart**, from one month's data. Naming which one you used, and why, is the skill.

DOMAINS 4, 9, 12

Why the number moved

Price/rate = (actual price − standard price) × actual quantity

price variance

usage/efficiency = (actual quantity − standard quantity) × standard price

usage variance — the other half of the story

EV = (points done / total points) × BAC

earned value when scope is a backlog

EMV = probability × impact

expected monetary value — how risk enters a forecast

PART ONE · 40%

Finance, accounting and reporting

1
5 KAs

Foundations of accounting for project controls

The accounting model · Components of the financial statements · Accrual accounting and the matching concept · Cost provisions and cost accruals · Chart of accounts and cost coding for projects

2
5 KAs

Financial reporting and the standards

The reporting framework · IFRS 15 Revenue from Contracts with Customers · Revenue recognition beyond the basics · Other relevant standards for project controls · Management reporting versus statutory reporting

3
5 KAs

Budgeting and forecasting

Budgeting fundamentals · Cost estimation · The time-phased budget, or cost baseline (planned value) · Forecasting · Cash-flow forecasting

4
4 KAs

Performance management, variance analysis and management reporting

Performance management principles · Variance analysis · Management reporting · Data visualisation and storytelling for controls

PART TWO · 40%

Project management

5
4 KAs

Cost management and cost control

The cost management framework · The cost control cycle · Cost breakdown and control accounts · Change control and cost impact

6
4 KAs

Earned value management and forecasting

EVM fundamentals · Variances and performance indices · Forecasting with EVM: the EAC family · Integrating cost and schedule, limitations, earned schedule

7
5 KAs

Contracts, commercial management, BoQ, invoicing and revenue

Types of contract · Contract management · Bills of quantities · Invoicing and applications for payment · Revenue recognition in the commercial cycle

8
6 KAs

Project management lifecycle

Initiating · Planning · Executing · Monitoring and controlling · Closing · Development approaches: predictive, iterative, incremental and adaptive

PART TWO · CONTINUED

Delivery, schedule, process and risk

9
6 KAs

Agile, Scrum and adaptive delivery for project controls

Agile foundations · The Scrum framework in depth · Backlogs, estimation and agile metrics · Kanban, Lean and scaling · Agile cost control, forecasting and earned value · Hybrid delivery and agile governance

10
4 KAs

Project scheduling, in depth

Schedule development · Network analysis and the critical path method · Schedule compression and resourcing · Progress measurement and schedule control

11
3 KAs

Business process cycles

Order-to-cash · Procure-to-pay · Internal control and segregation of duties

12
3 KAs

Risk management for project controls

The risk framework · The risk process · Contingency and management reserve

PART THREE · 20%

AI, governed

13
7 KAs

AI for project controls and project management

AI foundations for professionals · Data: the fuel · Prompting and working with generative AI · AI tool categories for project controls and PM · AI applied across the project-controls lifecycle · Governance, ethics, risk and assurance of AI · Building an AI-augmented project-controls capability

F1 = 2 × (precision × recall) ÷ (precision + recall)

whether the model is worth running at all

> A model with 99% recall and 4% precision buries the team in false positives. **"The AI found something" is not a control.**

BEHIND THE SYLLABUS

113 standards. 532 process requirements.

Each states its purpose, who owns the decision, what evidence must exist, what practice is prohibited, what triggers escalation, **what AI may never decide, approve or certify** — and a compliance test an assessor can perform.

They are certification requirements established by the Institute. They are not legislation, and nothing here is legal, tax or accounting advice. External standards are named and described in the Institute's own words, never reproduced.

THE PRINCIPLE BEHIND ALL OF IT

AI proposes. The professional verifies, decides and remains accountable.

The full outline and the three Bodies of Knowledge — projectcontrolsinstitute.org